UNIVERSITI PUTRA MALAYSIA
Bachelor of Computer Science with Honours

Forecasting Ethnic Composition Under Scenario-Based Migration Shocks Using Time-Series and NLP To Support Policy Planning

Name: Mohammad Javan Samboeputra Herlambang
Matric Number: 213878
Supervised by: Dr. Maslina Binti Zolkepli

Outline

- Introduction
- Objectives
- Scope
- Literature Review
- Methodology
- Implementation
- Result and Discussion
- Demo
- Conclusion

Introduction

The world is ever-changing—**conflict and migration** are constantly redrawing the map of human populations.

- These shifts disrupt **political dynamics, strain public services, challenge national identity,** and intensify debates around equity, inclusion, and representation.

Introduction

Ethnic composition—the proportion and distribution of different ethnic groups within a population (Wang, 2023) — is **rapidly changing** due to **migration, conflict, and climate shifts** (Pulami & Nepal, 2024).

- When governments fail to anticipate these changes, it can lead to **policy gaps in education, healthcare, and representation** (Dahinden & Groth, 2023).
- A better understanding of demographic volatility is essential for **inclusive and effective public planning.**

Problem statement

- Demographic forecasting often relies on **historical data and static models**, which **struggle to reflect sudden events**

(Dahinden & Groth, 2023; Drakaki & Tzionas, 2023)

- While NLP has been used to detect early migration signals, it remains **disconnected from time-series** forecasting

(Iliopoulos et al., 2022; Drakaki & Tzionas, 2023)

- Forecast results are often underutilized when **buried in static reports** or complex code outputs, **limiting their accessibility to policymakers and planners**

(Heggli et al., 2023)

Objectives

- To **adapt** machine learning-based **time-series models** for forecasting ethnic composition under **volatile geopolitical conditions**.

- To **apply** and **evaluate** the integration of **time-series** and **NLP-extracted data** to **produce insights for policy planning.**

- To **develop** an **interactive dashboard** that visualizes ethnic demographic forecasts and allows users to **explore scenario-based outcomes.**

Scope

- This project investigates how ethnic demographic forecasting can be **extended** by combining **structured census data** with **unstructured geopolitical news data**.

- The study focuses on a diverse set of countries that are **multicultural** by definition:

  the United States of America, Malaysia, Indonesia

- The project integrates NLP-based extraction of geopolitical signals from news with time-series model to **support scenario-based simulation of** ethnic shifts under real-world scenarios.

- Forecast results are visualized through an **interactive dashboard** to **facilitate exploration and interpretation** of how ethnic compositions may evolve under different geopolitical scenarios.

Literature review

An Investigation of Time Series Models for Forecasting Mixed Migration Flows: Focusing in Germany
Drakaki, M., & Tzionas, P. (2023).

- Doesn't apply NLP, Focused on Germany

National press monitoring using Natural Language Processing as an early warning signal for prediction of asylum applications flows in Europe
Iliopoulos, I., Kopalidis, N., Stavropoulos, G., & Tzovaras, D. (2022)

- Doesn't apply Time-Series, Focused in Europe

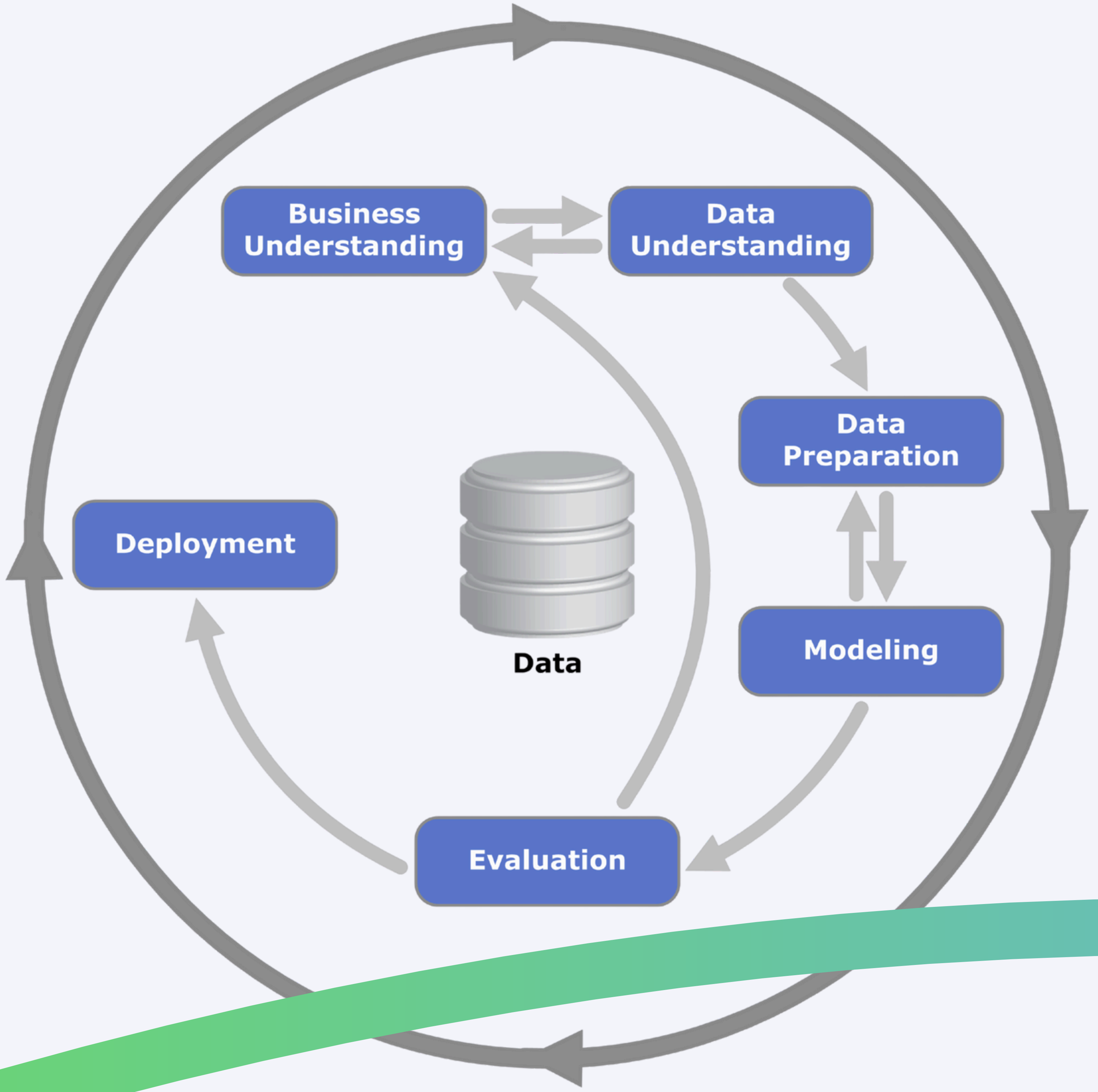
A Long Short-Term Memory Forecast of sub-National Population Change Across Europe
Newsham, N., Kumar, P., Cabrera-Arnau, C., & Rowe, F. (2024).

- Doesn't apply NLP, Focused in Europe

Feature	Drakaki & Tzionas (2023)	Kolambe & Arora (2024)	Iliopoulos et al. (2022)	Paiva et al. (2023)	Pekáček & Elmerot (2023)	Newsham et al. (2024)	This Project
Uses Time-Series Modeling	✅	✅	❌	❌	❌	✅	✅
Incorporates NLP	❌	❌	✅	✅	✅	❌	✅
Captures Time-Varying NLP Signals	❌	❌	❌	✅	½	❌	✅
Integrates Demographics with Geopolitical Signals	❌	❌	❌	½	❌	❌	✅
Highlights Ethnic Composition	❌	❌	❌	❌	❌	❌	✅
Supports Scenario-Based Policy Simulation	❌	❌	❌	❌	❌	❌	✅
Interactive Dashboard for Decision Support	❌	❌	❌	❌	❌	❌	✅

Methodology

CRISP-DM (Cross-Industry Standard Process for Data Mining)



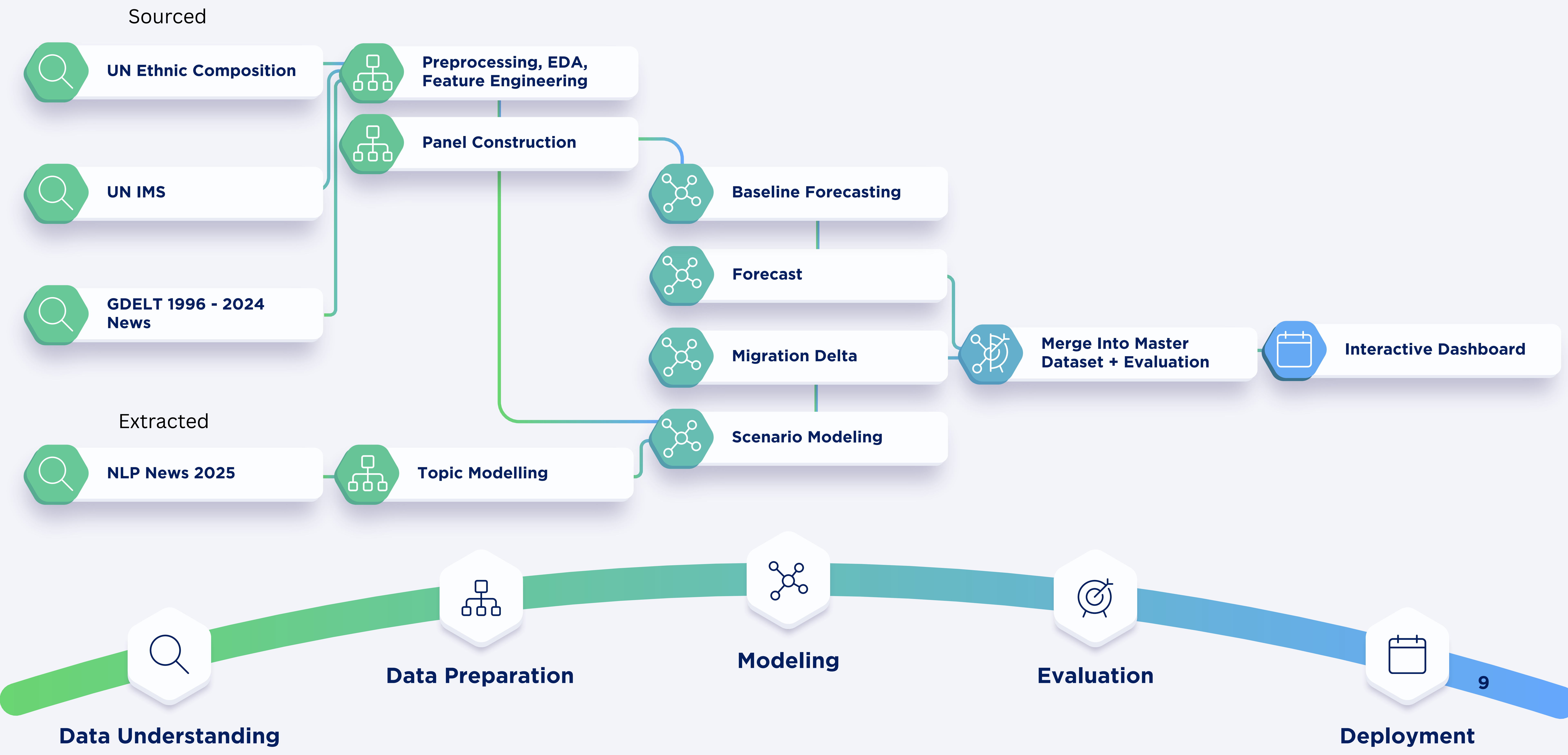
Established Objective:

Objectives

- To **adapt** machine learning-based **time-series models** for forecasting ethnic composition under **volatile geopolitical conditions**.
- To **apply** and **evaluate** the integration of **time-series** and **NLP-extracted data** to **produce insights for policy planning.**
- To **develop** an **interactive dashboard** that visualizes ethnic demographic forecasts and allows users to **explore scenario-based outcomes.**

Business Understanding

Methodology - Flowchart



Implementation

1.1-Data Preprocessing.ipynb Libraries: pandas, numpy, country_converter

Ethnic Composition Data

...untry or	Year	Area	Sex	National and/or eth	Record Ty	Reliability	Source Ye	Value	Value Footnote
Ã...land Is	2000	Total	Both Sexes	Total	Census - c	Final figur	2009	25776	1
Ã...land Is	2000	Total	Both Sexes	Finnish	Census - c	Final figur	2009	5109	1
Ã...land Is	2000	Total	Both Sexes	Swedish	Census - c	Final figur	2009	1354	1
Ã...land Is	2000	Total	Both Sexes	Other	Census - c	Final figur	2009	552	1
Ã...land Is	2000	Total	Both Sexes	Ã...land	Census - c	Final figur	2009	18682	1
Ã...land Is	2000	Total	Both Sexes	Other Nordic Countr	Census - c	Final figur	2009	79	1
Ã...land Is	2000	Total	Male	Total	Census - c	Final figur	2009	12700	1
...land Is	2000	Total	Male	Finnish	Census - c	Final figur	2009	2027	1

International Migrant Stock Data

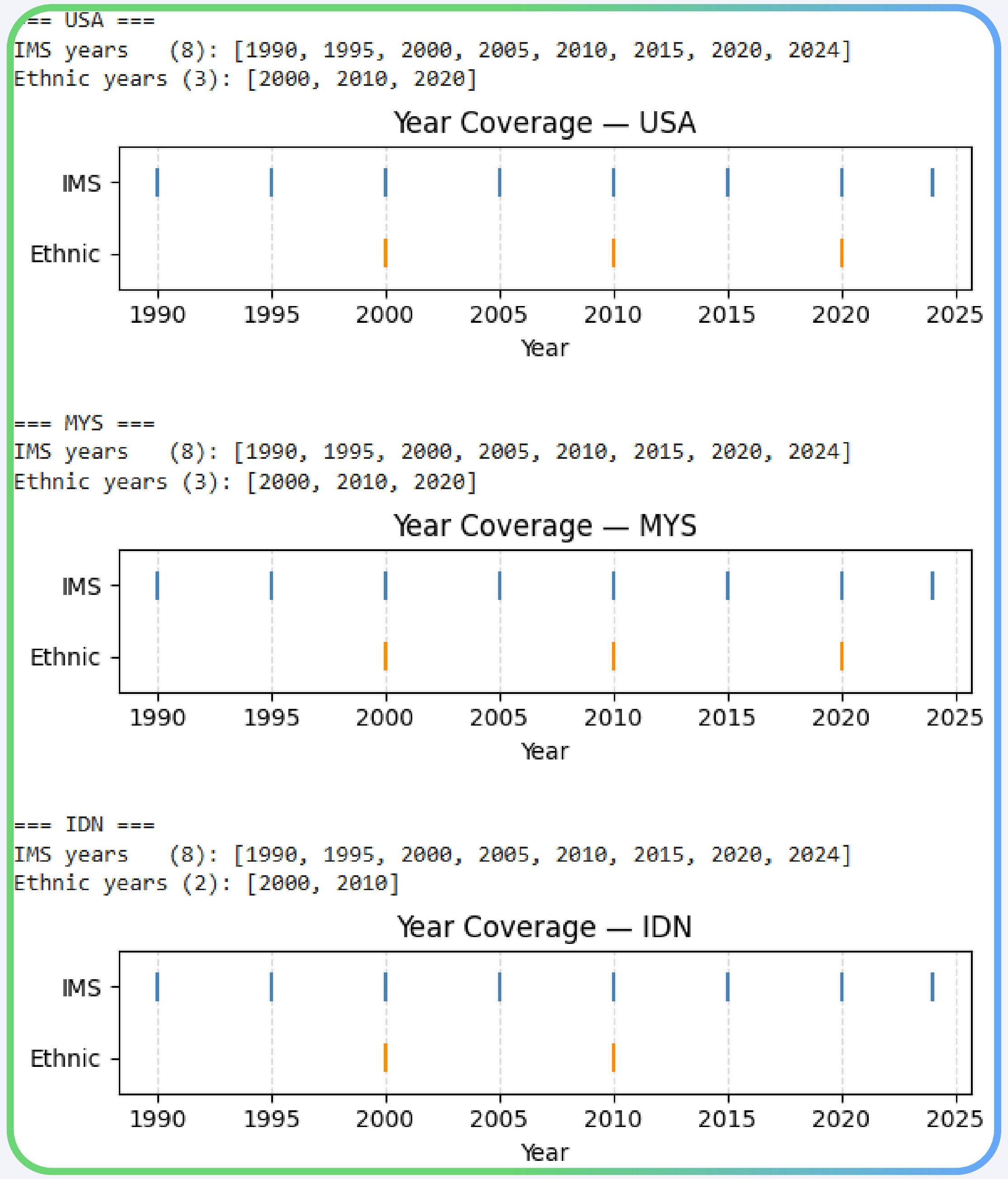
	Region, development group, country or area	Location code	International migrant sto... 1990	1995	2000
1	World	900	153 916 063	163 176 002	174 566 152
2	Sub-Saharan Africa	1834	13 802 710	14 578 211	13 518 102
3	Northern Africa and Western Asia	1833	16 863 681	18 573 252	20 033 534
4	Central and Southern Asia	1831	25 456 471	20 140 424	18 979 217
5	Eastern and South-Eastern Asia	1832	6 755 128	8 203 094	10 530 525
6	Latin America and the Caribbean	1830	7 253 323	6 728 705	6 589 509
7	Oceania (excluding Australia and New Zealand)	1835	258 678	280 580	296 618
8	Australia/New Zealand	1836	4 517 870	4 810 557	5 075 813
9	Europe and Northern America	1829	79 008 202	89 861 179	99 542 83...

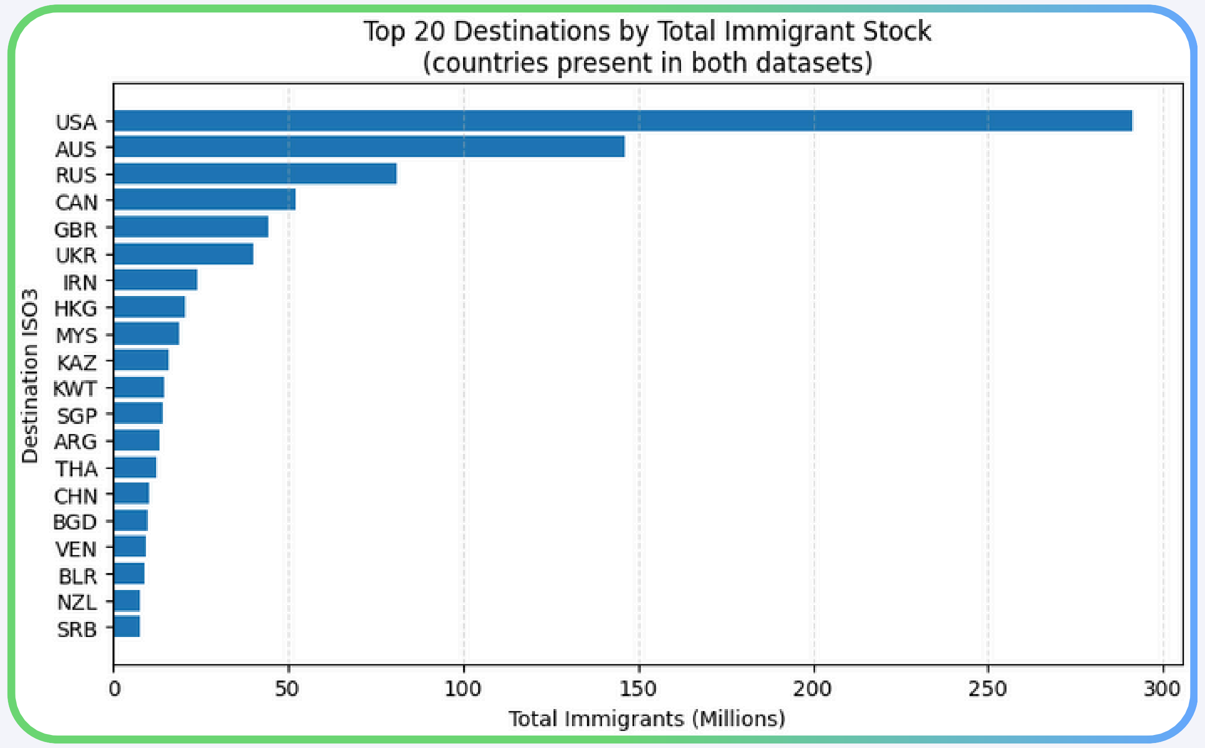
- Cleaned and standardized ethnic and IMS data
  - Ethnic: Removed **Footnotes** and **header repetitions**, filtered to **total population** and **both sexes**, corrected **mojibake.**
  - IMS: Filtered out aggregate regions (like 'World', 'Europe'), transformed the wide-format year columns into a long format.
- Harmonized country identifiers using ISO3 codes
- Identified **origin–destination** relationships

```
A…land Islands": "Aland Islands",
CÃ´te dâ€™Ivoire": "Côte d’Ivoire",
CuraÃ§ao": "Curaçao",
RÃ©union": "Réunion",
SÃ£o TomÃ© and PrÃncipe": "São Tomé and Príncipe"
```

Implementation

1.2-Exploratory Data Analysis.ipynb Libraries: pandas, numpy, matplotlib, seaborn, country_converter

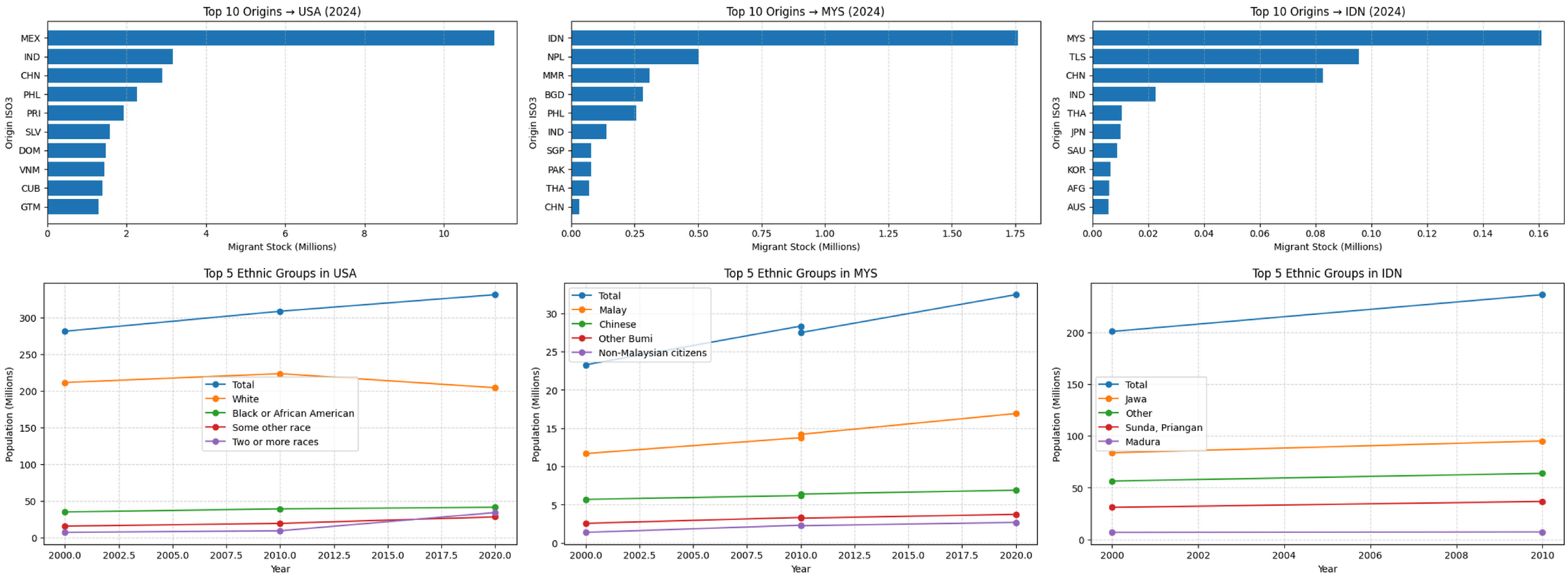




- Assessed **data coverage** and **temporal consistency**
- Identified **dominant ethnic groups** and **migration patterns**
- Validated trends across selected countries

Implementation

1.2-Exploratory Data Analysis.ipynb Libraries: pandas, numpy, matplotlib, seaborn, country_converter



Implementation

1.3-Feature Engineering.ipynb Libraries: pandas, numpy

Ethnic Composition Data

International Migrant Stock Data

```
needed_iso3 = [
    'AFG','ARG','AUS','BGD','BRA','BRN','CAN','CHN','COL','CUB','DEU','DOM','ECU','EGY','ERI','ESP','ETH','FRA','GBR',
    'GHA','GRC','GTM','GUY','HKG','HND','HTI','IDN','IND','IRL','IRN','IRQ','ISR','ITA','JAM','JPN','KEN','KHM','KOR',
    'LAO','LBN','LKA','MEX','MMR','MYS','NGA','NIC','NPL','NZL','PAK','PAN','PER','PHL','POL','PRI','PRT','PSE','ROU',
    'RUS','SAU','SDN','SGP','SLV','SOM','SYR','THA','TLS','TTO','TUR','UKR','USA','VEN','VNM','YEM'
]
```

- Computed ethnic population shares and totals
  - Ethnic: **Filtering** relevant countries, removing **'Total' groups** (per country), computing total populations, and calculating ethnic shares.
  - IMS: **Filter** for specific 'core' countries (USA, MYS, IDN) as both **origins and destinations**, and calculates **migrant stocks** and their **shares**.
- Interpolated values to create continuous time series (**2000, 2005, 2010, 2015, 2020**)
- Simple forecasting on 2024 using **linear regression**

	iso3	ethnic_group	year	share_interp
0	ARG	African	2000	0.005030
1	ARG	African	2005	0.005030
2	ARG	African	2010	0.005030
3	ARG	African	2015	0.005030
4	ARG	African	2020	0.005030
...	...	...	...	...
10475	VNM	Xtieng	2000	0.000915
10476	VNM	Xtieng	2005	0.000955
10477	VNM	Xtieng	2010	0.001013
10478	VNM	Xtieng	2015	0.001030
10479	VNM	Xtieng	2020	0.001047

Implementation

CO 2.1-GDELT.ipynb BigQuery, Libraries: pandas, numpy, pycountry

```
H categorized AS (
SELECT
  ActionGeo_CountryCode AS country,
  Year AS year,
  CASE
    WHEN EventRootCode IN ('18','19','20') THEN 'Conflict/Migration'
    WHEN EventRootCode = '13' THEN 'Protest'
    WHEN EventRootCode IN ('04','05') THEN 'Aid/Cooperation'
    WHEN EventRootCode BETWEEN '10' AND '12' THEN 'Disapproval/Sanction'
    ELSE 'Other'
  END AS category,
  GoldsteinScale AS goldstein,
  AvgTone AS tone
FROM `gdelt-bq.full.events`
WHERE
  Year BETWEEN 1996 AND 2024
  AND ActionGeo_CountryCode IN (
    'AF','AR','AU','BD','BR','BN','CA','CN','CO','CU','DE',
    'DO','EC','EG','ER','ES','ET','FR','GB','GH','GR','GT',
    'GY','HK','HN','HT','ID','IN','IE','IR','IQ','IL','IT',
    'JM','JP','KE','KH','KR','LA','LB','LK','MX','MM','MY',
    'NG','NI','NP','NZ','PK','PA','PE','PH','PL','PR','PT',
    'PS','RO','RU','SA','SD','SG','SV','SO','SY','TH','TL',
    'TT','TR','UA','US','VE','VN','YE'
  )


ELECT
 country,
 year,
 category,
 COUNT(*)                AS event_count,
 ROUND(AVG(goldstein), 3)   AS goldstein_avg,
 ROUND(AVG(tone), 3)        AS tone_avg
ROM categorized
ROUP BY country, year, category
DER BY country, year, category;
```

gdelt_country_year_category.csv

A	B	C	D	E	F
country	year	category	event_count	goldstein_avg	tone_avg
AF	1996	Aid/Cooperation	11059	3.429	5.219
AF	1996	Conflict/Migration	4606	-9.875	3.322
AF	1996	Disapproval/Sanctior	2269	-3.15	4.703
AF	1996	Other	9605	2.027	4.869
AF	1996	Protest	473	-5.585	4.402
AF	1997	Aid/Cooperation	10718	3.292	5.386
AF	1997	Conflict/Migration	4465	-9.868	3.65

- Extracted geopolitical event indicators from **GDELT's BigQuery** ('Conflict/Migration', 'Protest', 'Aid', and 'Disapproval'.)
- Grouped data into **'hybrid windows'** to match Ethnic and IMS datasets. (e.g., 2001-2005 maps to 2005)
- Aggregated events into **country-year** metrics
- Derived conflict and cooperation indicators

Implementation

2.2-NLP Topics.ipynb api.gdeltproject.org/api/v2/doc/doc

```
429 rate limit 2025-10-10-2025-10-13 | sleeping 2.6s (attempt 1)
429 rate limit 2025-11-03-2025-11-06 | sleeping 2.1s (attempt 1)
429 rate limit 2025-11-27-2025-11-30 | sleeping 2.4s (attempt 1)
429 rate limit 2025-12-21-2025-12-24 | sleeping 2.4s (attempt 1)
429 rate limit 2025-12-27-2025-12-30 | sleeping 2.9s (attempt 1)
Global news shape: (87250, 10)
topic_seed counts:
topic_seed
conflict     29250
migration    29000
coop         29000
Name: count, dtype: int64
datetime range:
2025-01-01 00:15:00+00:00 → 2025-12-31 00:15:00+00:00
```

	url	url_mobile	title	seendate	socialimage	domain
0	https://allafrica.com/stories/202501020200.html		Nigeria : 7 African Countries On the U . S . G...	20250102T143000Z	https://cdn01.allafrica.com/download/pic/main/...	allafrica.com
1	https://www.prabhatkhabar.com/opinion/children...	https://www.prabhatkhabar.com/opinion/children...	हिंसा का शिकार होते बच्चे I children victims o...	20250102T063000Z	https://www.prabhatkhabar.com/wp-content/uploa...	prabhatkhabar.com
2	https://www.elimparcial.com/mundo/2025/01/01/f...	https://www.elimparcial.com/mundo/2025/01/01/f...	Fallecimiento de deportistas en 2024 : Estas f...	20250101T201500Z	https://www.elimparcial.com/resizer/v2/NOGS3Y3...	elimparcial.com
3	https://www.africanews.com/2025/01/01/sudans-l...	/amp/2025/01/01/sudans-leader-rules-out-pre-wa...	Sudan leader rules out pre - war reconciliatio...	20250101T134500Z	https://static.euronews.com/articles/stories/0...	africanews.com
4	https://www.aljazeera.net/news/2025/1/1/%D9%87...	https://www.aljazeera.net/amp/news/2025/1/1/%d...	هجوم على كييف برأس السنة وزيلينسكي يقدم رؤيته ...	20250101T111500Z	https://www.aljazeera.net/wp-content/uploads/2...	aljazeera.net

```
aliases = {
    # AFG
    "afghan": "AFG",
    "afghanistan": "AFG",
    "افغانستان": "AFG",          # Arabic / Persian / Urdu
    "افغان": "AFG",

    # ARG
    "argentina": "ARG",
    "аргентина": "ARG",           # Russian / Ukrainian
    "argentinien": "ARG",         # German
    "argentine": "ARG",           # French

    # AUS
    "australia": "AUS",
    "australie": "AUS",           # French
    "australien": "AUS",          # German
    "австралия": "AUS",           # Russian
    "澳大利亚": "AUS",            # Chinese (simplified)
    "澳大利亞": "AUS",            # Chinese (traditional)

    # BGD
    "bangladesh": "BGD",
    "bangla": "BGD",
    "বাংলাদেশ": "BGD",            # Bengali
```

- Identified countries in **news titles** and categorized them as 'destination' or 'origin'
- Categorized countries as 'destination' or 'origin'.
- Applied topic modeling (**BERTopic**) to migration-related news articles
  - Classified topics into conflict, migration, and cooperation
- Applied targeted **sentiment analysis** on topics
- Aggregated signals for scenario analysis,
  - set volume of articles as **intensity factor**

Implementation

3.1-Final Panel Builder 2.0.ipynb Libraries: pandas, numpy, pycountry

Load ethnic composition (base)

↳ 2 cells hidden

Load IMS destination and origin features (

↳ 3 cells hidden

Load GDELT

↳ 2 cells hidden

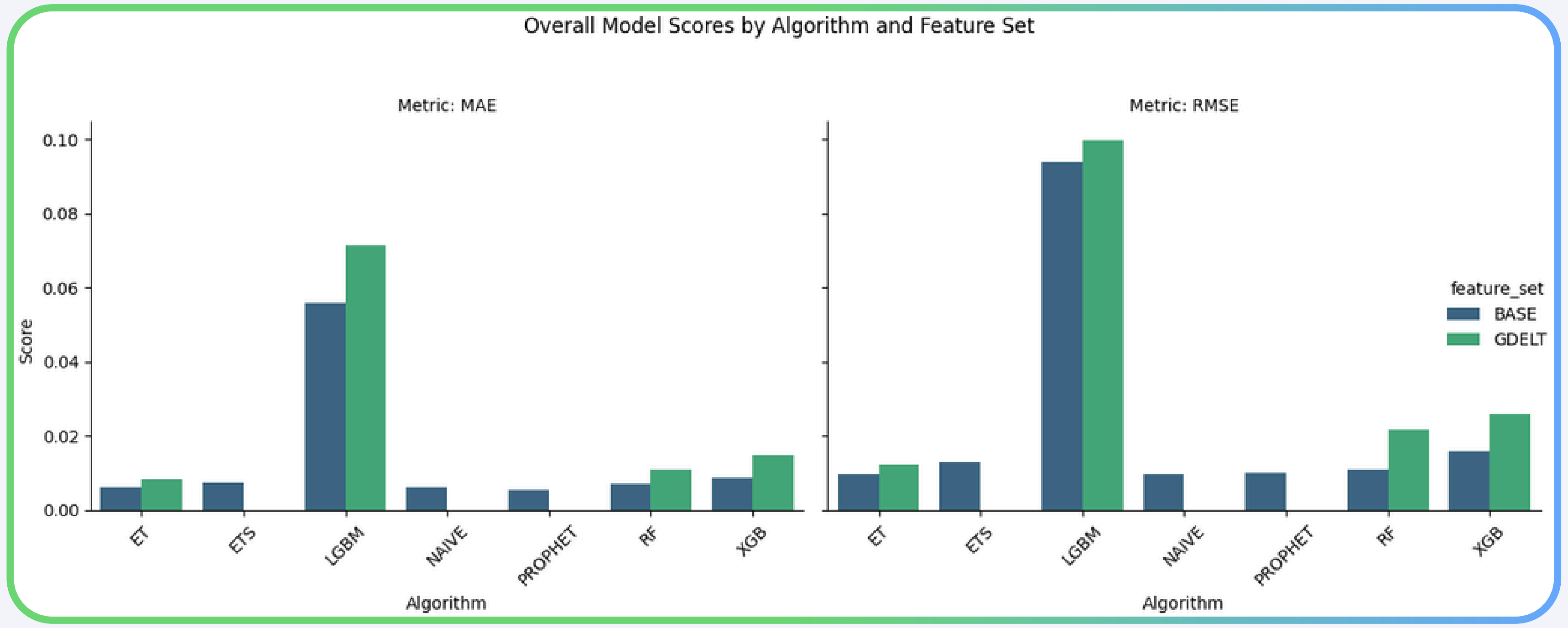
share_interp_lag1	share_interp_lag2
0.01738641449	
0.02874260529	0.01738641449
0.0400987961	0.02874260529
0.0400987961	0.0400987961
0.0400987961	0.040098796

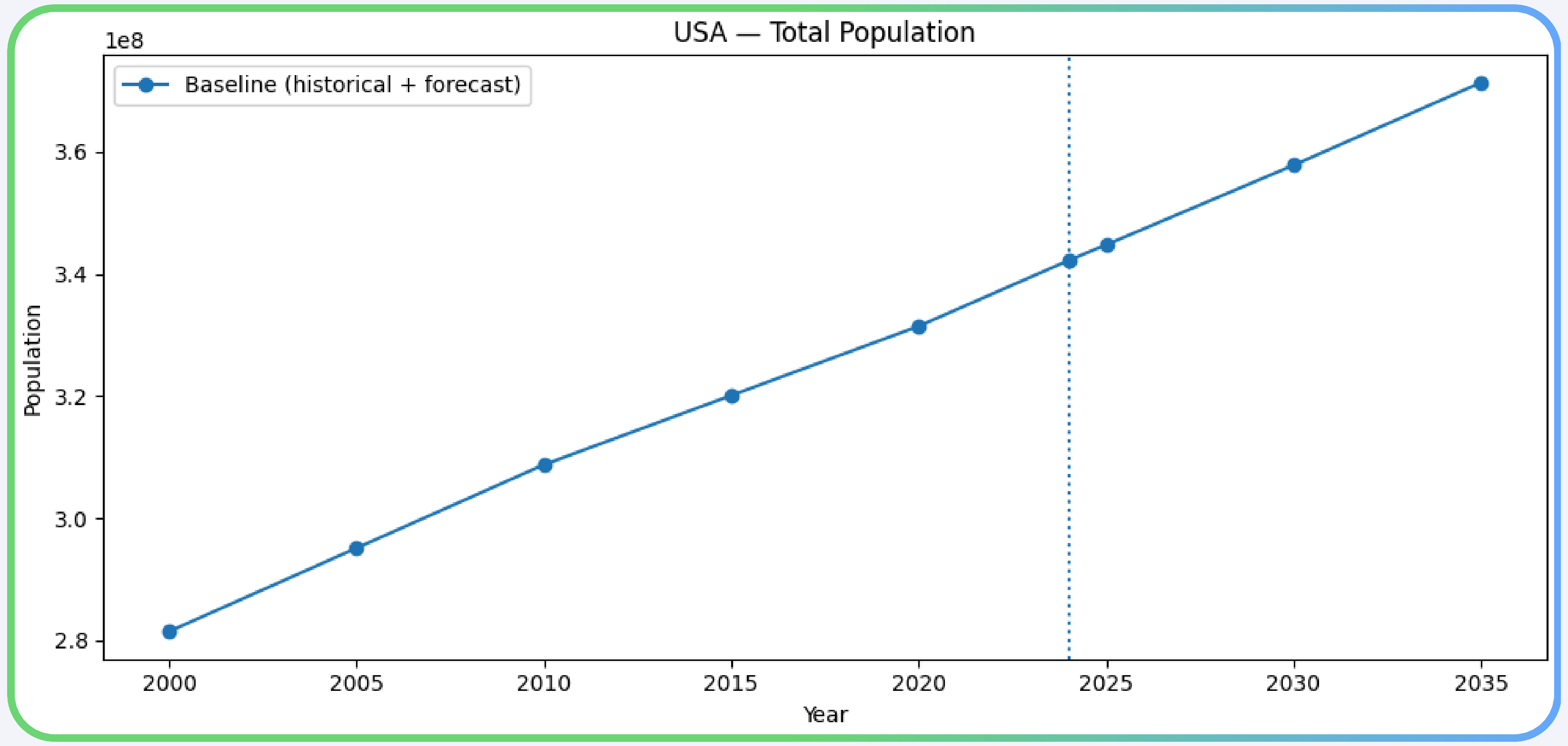
```
# Merge origin gdelt onto flows (per origin-year)
flows_g = flows.merge(
    gdelt_orig,
    left_on=["iso3_orig", "year"],
    right_on=["orig_iso3", "year"],
    how="left"
)

# Weighted aggregation: for each dest-year,
# compute weighted mean of origin gdelt signals
# weight = migrant_stock
w = flows_g[weight_col].astype(float).replace(0.0, np.nan)
```

- Constructed a core panel by integrating **ethnic composition** data with **IMS** features, including **lagged ethnic shares**.
- Aggregated **geopolitical signals from GDELT** for **destination** and **origin** countries, weighted by migration flows.
- Merged **geopolitical features** with the **core panel** to form an enriched panel.
- Derived **country-level migration pressure** indicators by combining conflict and cooperation signals into a **unified delta driver** for scenario modeling.

Implementation

3.2-Modeling 2.0.ipynb Libraries: pandas, numpy, pycountry



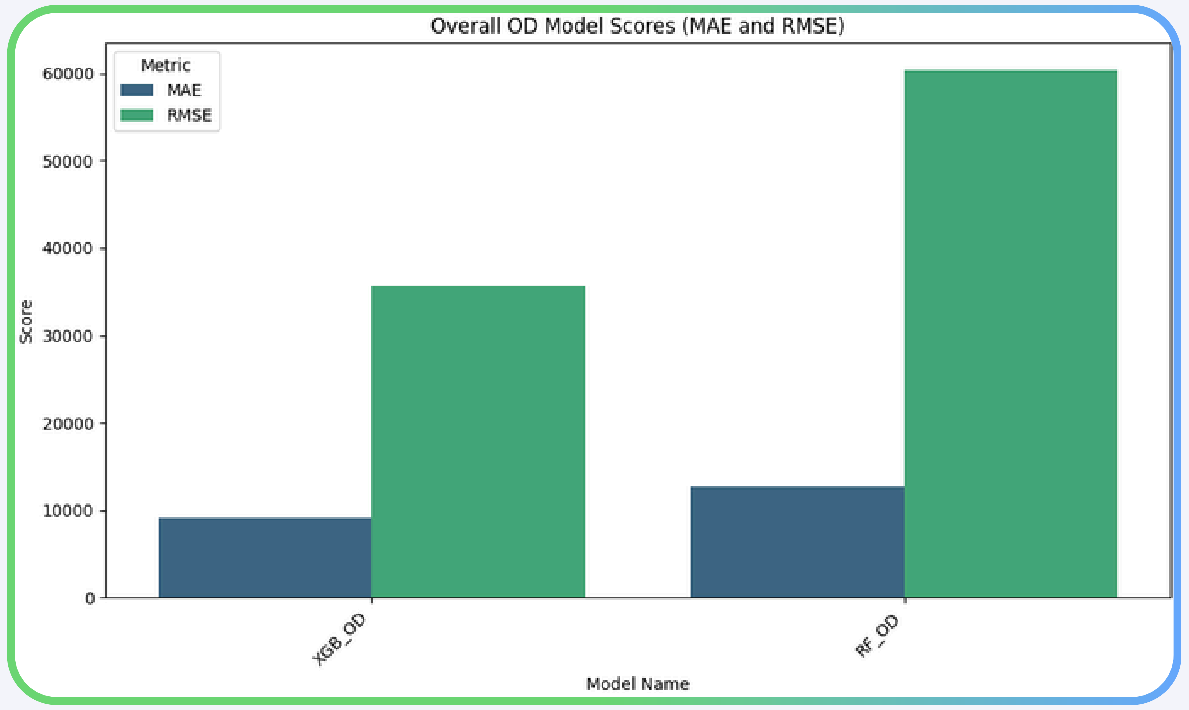


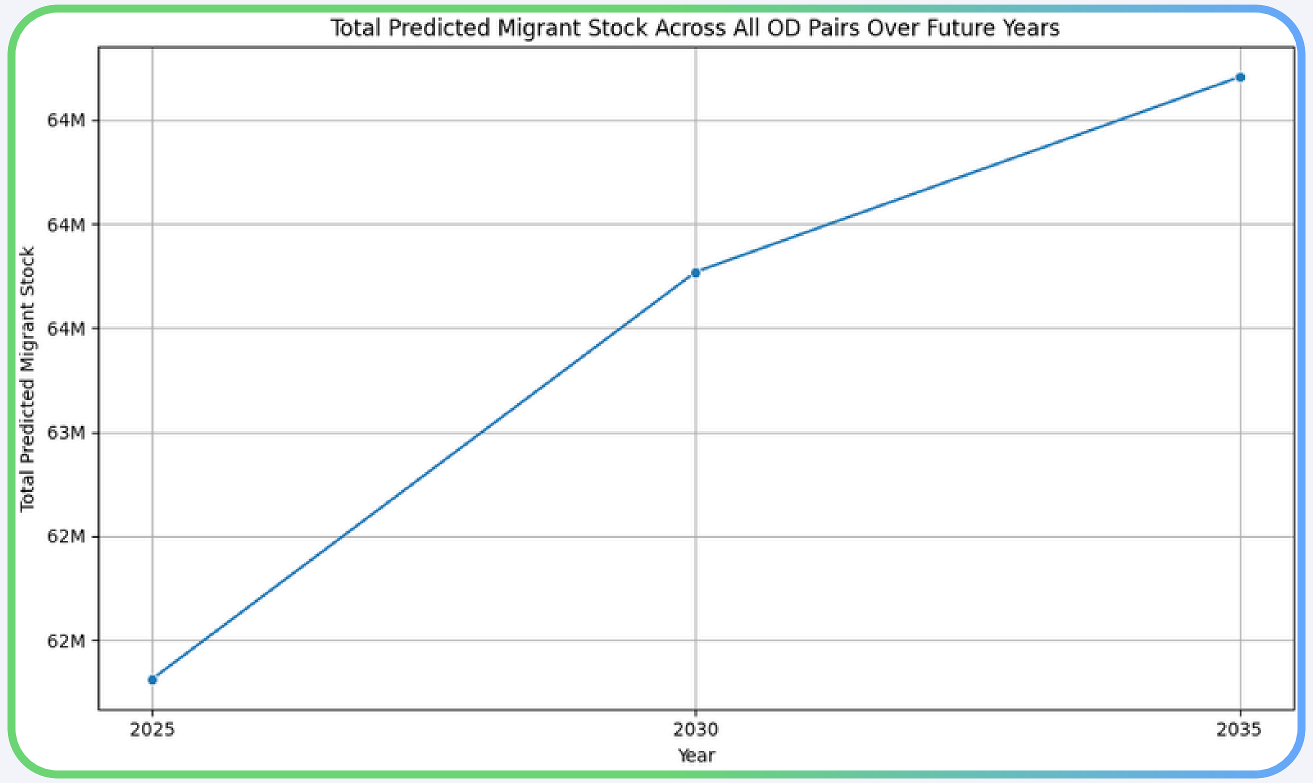
Baseline Modeling

- Conducted **model evaluation** using a **2024 holdout**, comparing naïve and classical baselines with multiple machine learning models under **BASE** and **GDELT** feature sets.
- Selected the best-performing models;
  Best baseline: **ET_BASE**, Best scenario: **ET_GDELT**
- Generated multi-year ethnic composition forecasts with **normalization** and **anchoring**.

Implementation

3.2-Modeling 2.0.ipynb Libraries: pandas, numpy, pycountry



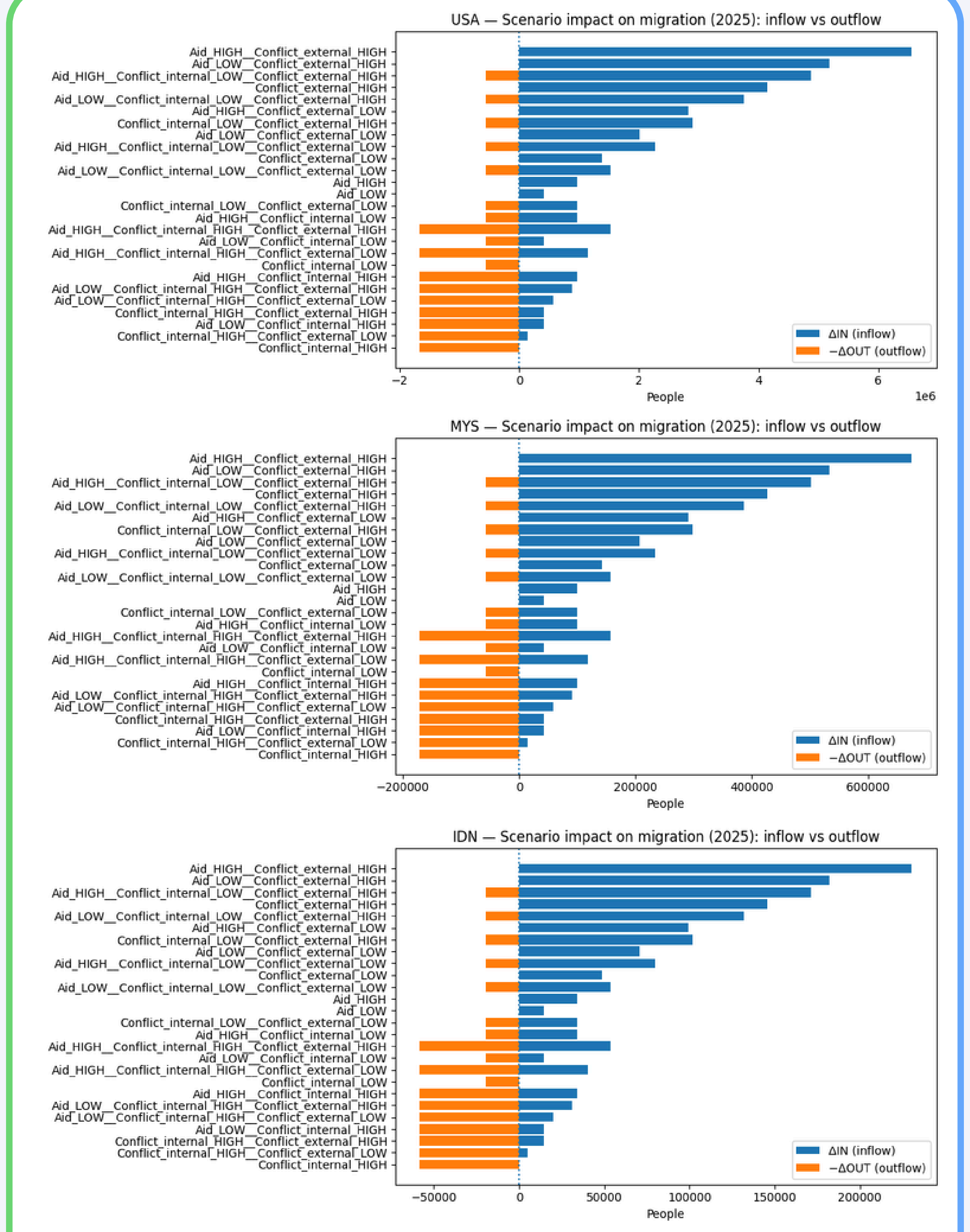


Scenario Modeling

- Performed **Origin-Destination (OD)** flow forecasting.
- Engineered **lag features** for migrant stock.
- Splitting data, training **RandomForest** and **XGBoost** models, evaluating their performance, and generating **forecasts** for future years.

Implementation

3.3-Migration Delta 2.0.ipynb Libraries: pandas, numpy, pycountry



- Applied migration shock scenarios using **NLP-based geopolitical drivers** for 2025

- Simulated **migration deltas** under multiple scenarios:
  - Aid Low/High
  - Conflict Internal Low/High
  - Conflict External Low/High

- Distributed scenario migration deltas to OD flows using historical weights.

- Mapped OD shocks into **destination ethnic buckets**.

Implementation

4.0-build_master_dataset.ipynb BigQuery, Libraries: pandas, numpy, pycountry

	iso3	year	ethnic_group	share	count	total_pop	model_name	view_type	scenario_name	delta_migrants	delta_in	delta_out
0	IDN	2000	Banjar, Melayu Banjar	0.017386	3.496273e+06	2.010922e+08	HIST	HIST	NaN	NaN	NaN	NaN
1	IDN	2000	Banten	0.020454	4.113162e+06	2.010922e+08	HIST	HIST	NaN	NaN	NaN	NaN
2	IDN	2000	Betawi	0.025072	5.041688e+06	2.010922e+08	HIST	HIST	NaN	NaN	NaN	NaN
3	IDN	2000	Bugis, Ugi	0.024916	5.010423e+06	2.010922e+08	HIST	HIST	NaN	NaN	NaN	NaN
4	IDN	2000	Jawa	0.416490	8.375285e+07	2.010922e+08	HIST	HIST	NaN	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...
817	USA	2035	Black or African American	0.109745	4.075452e+07	3.713570e+08	ET_BASE	FORECAST	NaN	NaN	NaN	NaN
818	USA	2035	Native Hawaiian and other Pacific Islander	0.077876	2.891977e+07	3.713570e+08	ET_BASE	FORECAST	NaN	NaN	NaN	NaN
819	USA	2035	Some other race	0.094166	3.496909e+07	3.713570e+08	ET_BASE	FORECAST	NaN	NaN	NaN	NaN
820	USA	2035	Two or more races	0.101408	3.765860e+07	3.713570e+08	ET_BASE	FORECAST	NaN	NaN	NaN	NaN
821	USA	2035	White	0.447312	1.661125e+08	3.713570e+08	ET_BASE	FORECAST	NaN	NaN	NaN	NaN

822 rows × 17 columns

	iso3_orig	iso3_dest	year	migrant_stock	migrant_stock_pred	migrant_stock_shock	delta_in_od	delta_out_od
0	AFG	IDN	2000	78.0	NaN	NaN	NaN	NaN
1	AFG	IDN	2005	36.0	NaN	NaN	NaN	NaN
2	AFG	IDN	2010	1548.0	NaN	NaN	NaN	NaN
3	AFG	IDN	2015	6263.0	NaN	NaN	NaN	NaN
4	AFG	IDN	2020	7646.0	NaN	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...
3845	VNM	USA	2025	NaN	1.431834e+06	1.431834e+06	0.000000	0.0
3846	VNM	USA	2025	NaN	1.431834e+06	1.517308e+06	85474.299543	0.0
3847	VNM	USA	2025	NaN				
3848	VNM	USA	2030	NaN				
3849	VNM	USA	2035	NaN				

iso3_dest	iso3_orig	origin_ethnic_group	dest_ethnic_bucket
IDN	AFG		Other
IDN	AUS	Acehnese	Other
IDN	AUS	Acholi	Other
IDN	AUS	Afgan	Other
IDN	AUS	African American	Other
IDN	AUS	African, so describec	Other
IDN	AUS	Afrikaner	Other

Ethnic Demography Master Data

- Combined historical, forecast, and scenario ethnic data into one master dataset.
- Ensured consistency of ethnic shares and population counts.

OD Migration Master Data

- Combined historical, forecast, and scenario OD migration data into a single dataset.

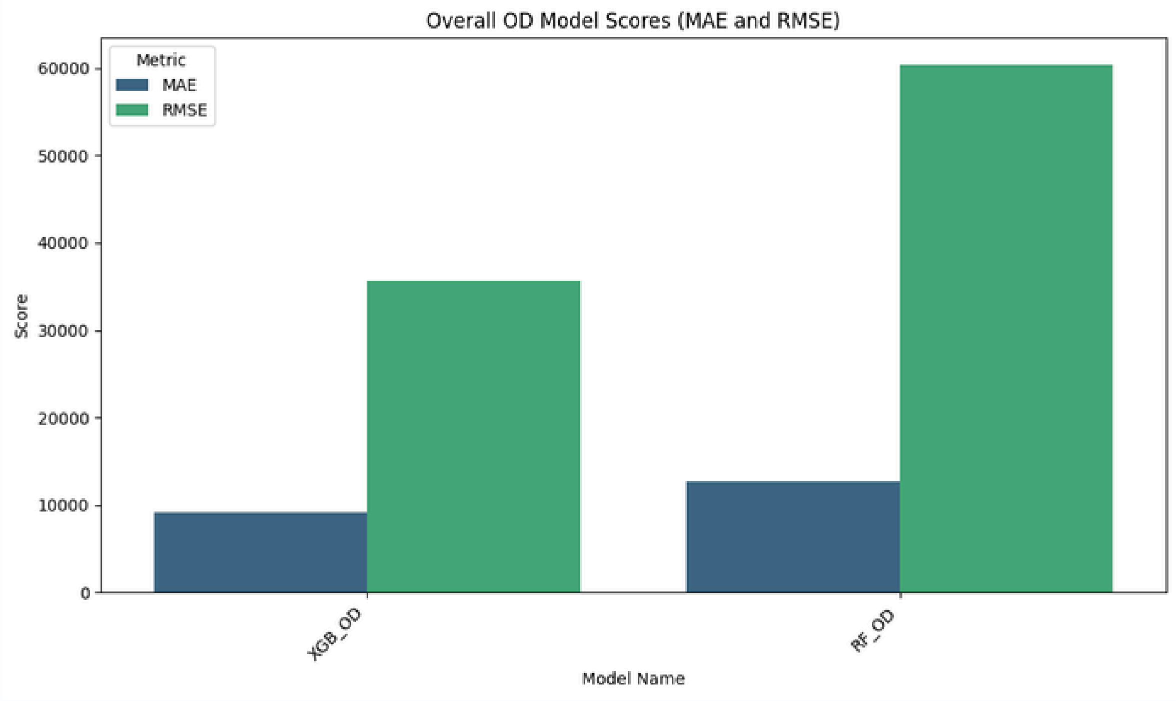
OD Impact: Allocation

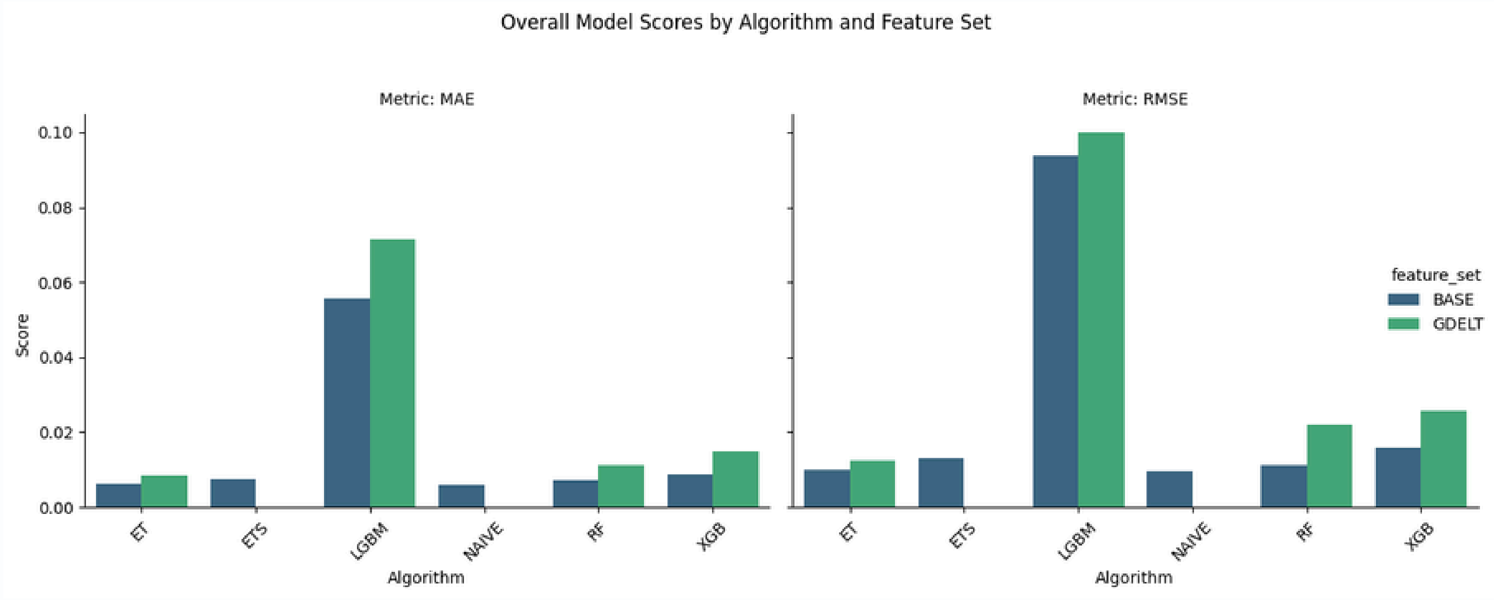
- Allocated scenario migration impacts into destination ethnic groups.
- Prepared aggregated outputs for analysis and visualization.

Results and Discussion

Result 1 — Model Performance Validation

- Machine learning models performs similar to baselines
- Stable error across ethnic groups and countries
- Selected model used for final forecasts

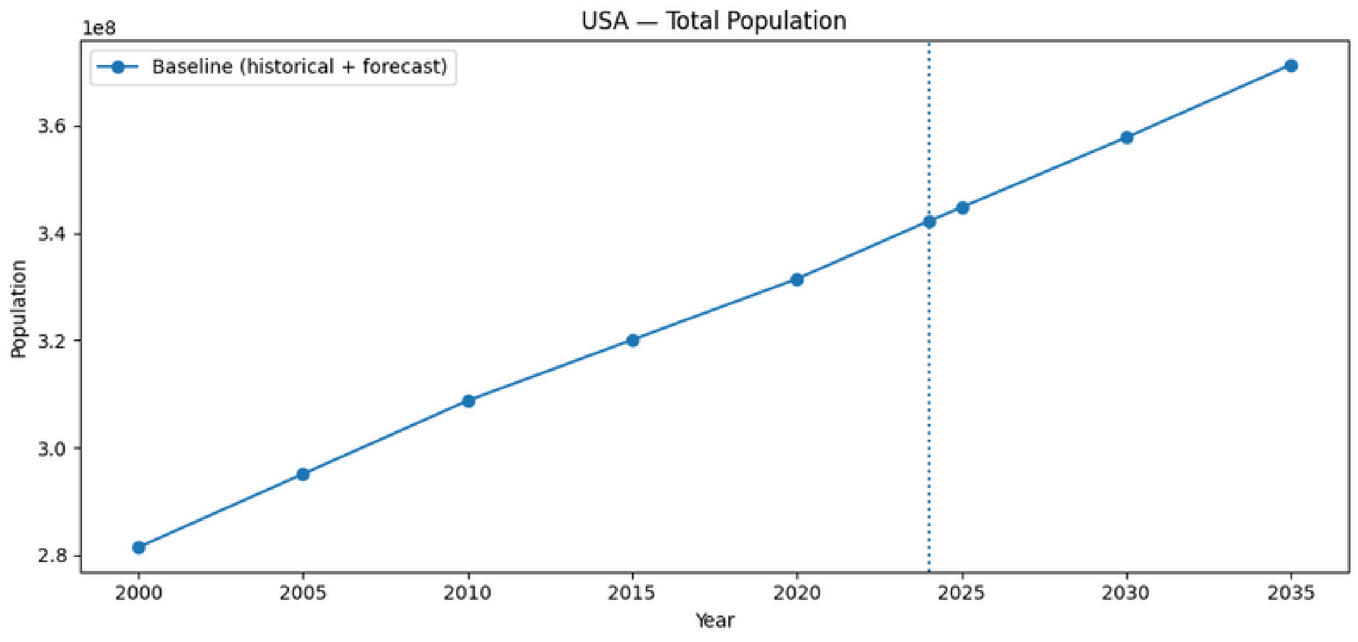


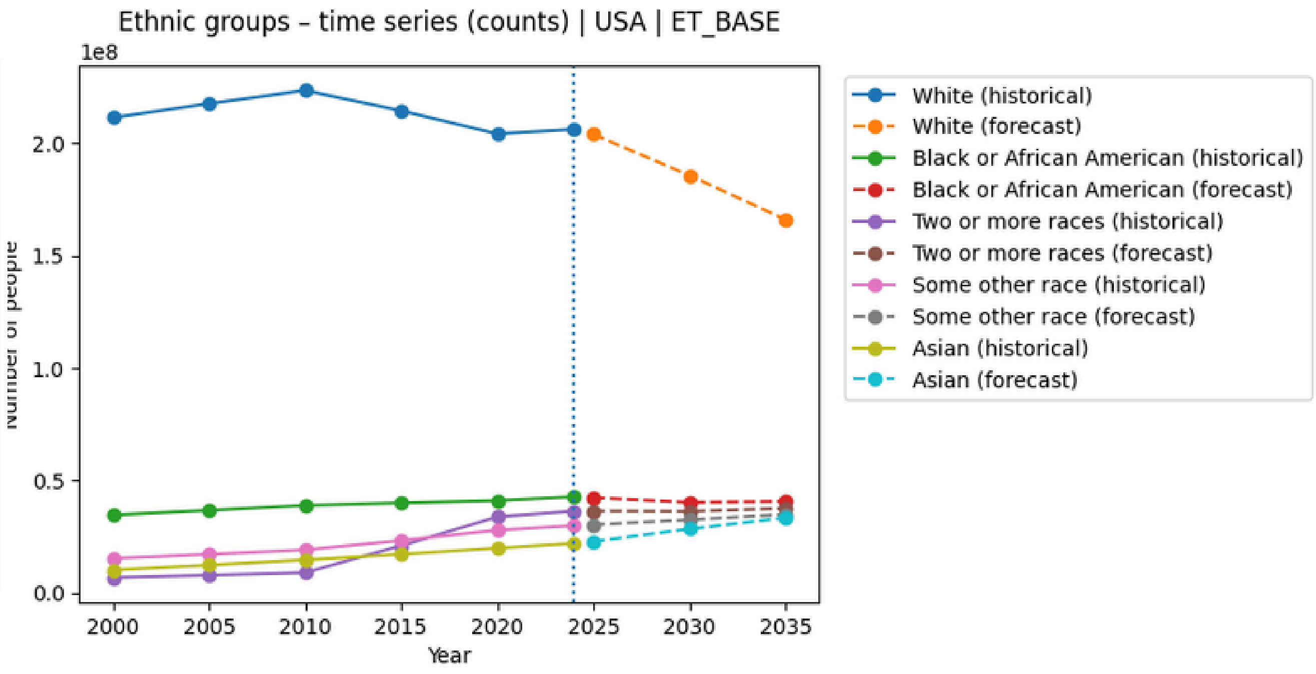


Results and Discussion

Result 2 — Baseline Ethnic Forecast

- Forecasted ethnic composition for 2025–2035
- Smooth trend continuation under baseline assumptions
- Shares remain normalized and interpretable

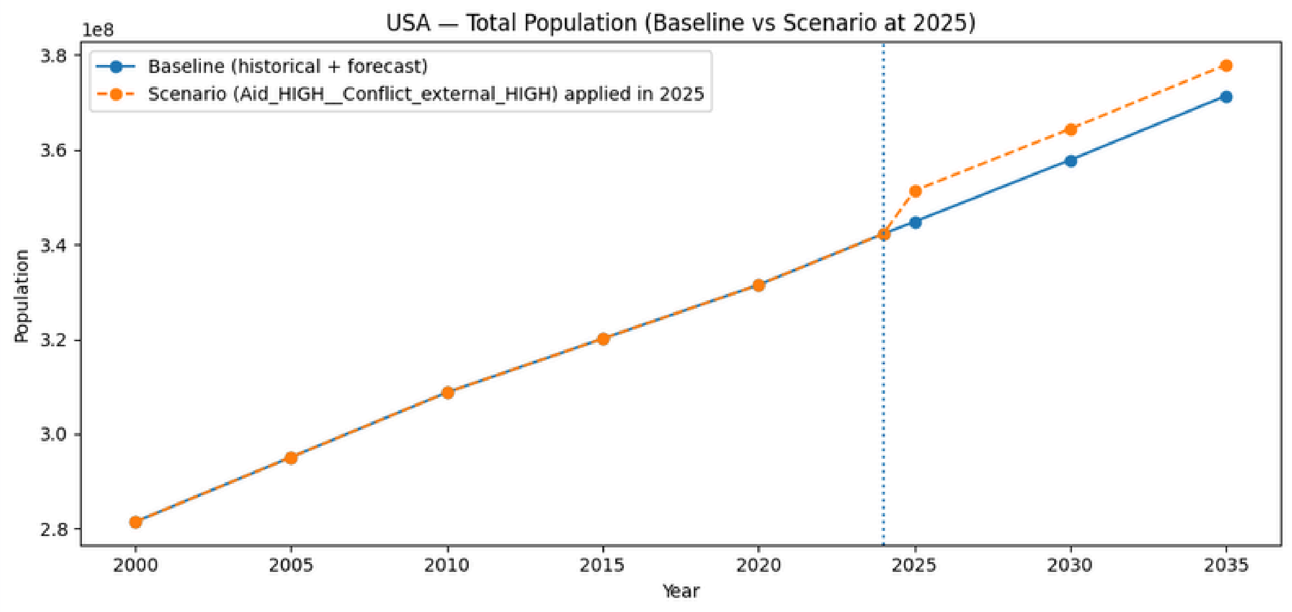


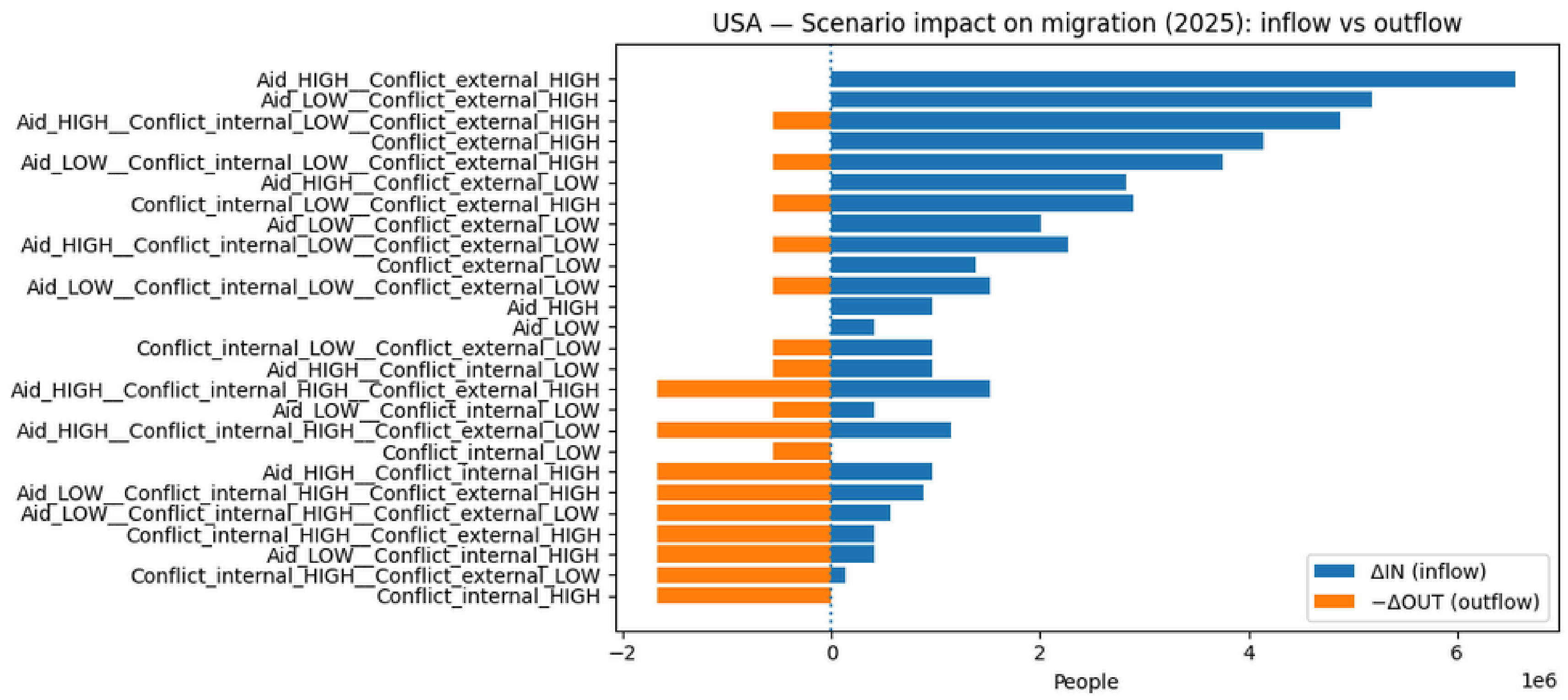


Results and Discussion

Result 3 — Scenario Impact Analysis

- Geopolitical shocks alter ethnic composition outcomes
- Migration-driven changes are uneven across groups
- Scenarios reveal non-linear demographic effects

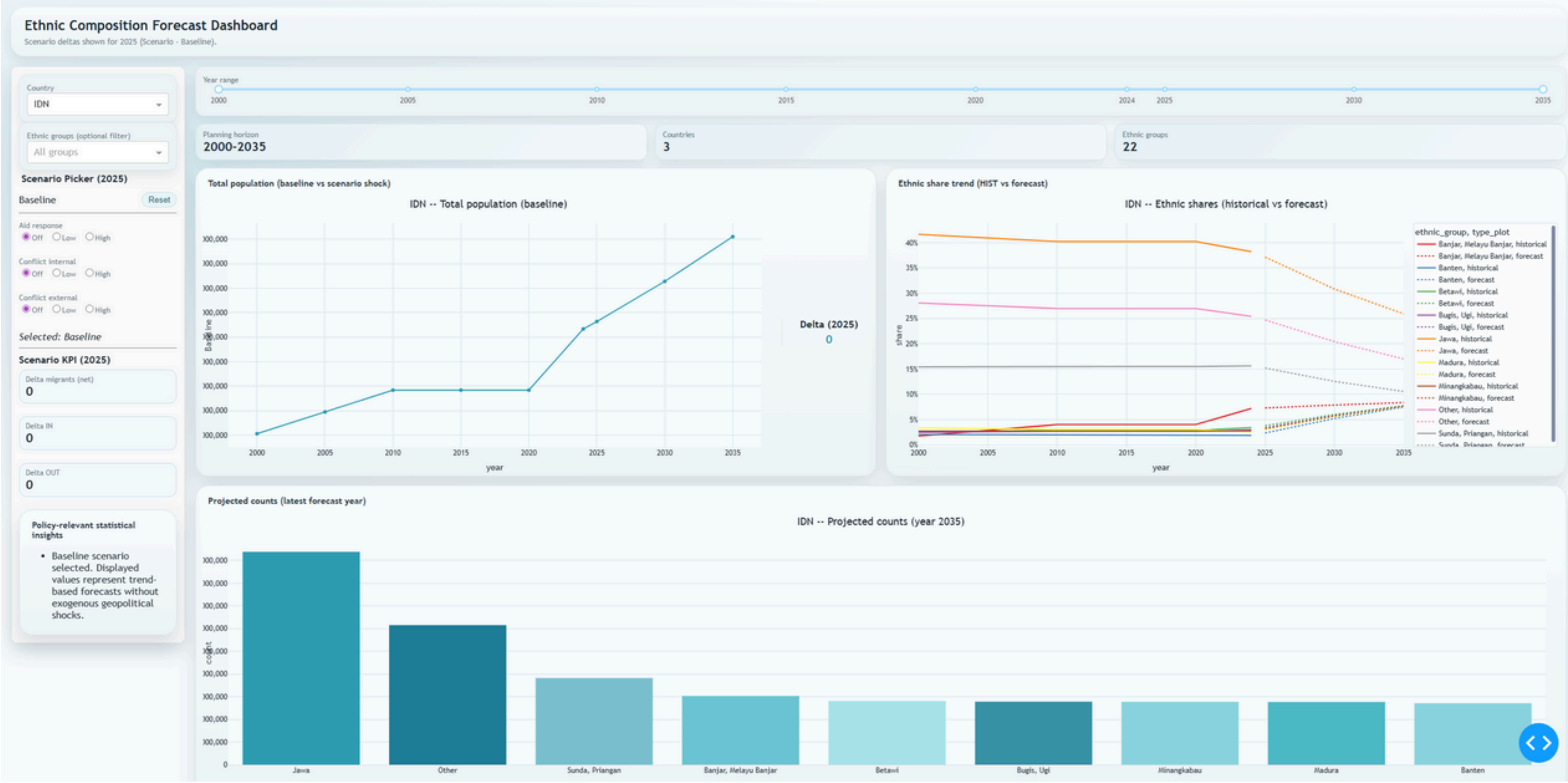




Results and Discussion

Result 4 — Dashboard Output

- Interactive exploration of forecasts and scenarios
- Designed for non-technical policy users
- Supports early warning and planning use cases



Demo

Conclusion

- Ethnic composition forecasting benefits from **integrating migration and geopolitical context**
- Time-series models provide **stable baselines for demographic trends**
- NLP-derived signals add value through scenario-based analysis rather than direct prediction
- The system demonstrates how data-driven tools can support anticipatory policy planning

Conclusion - Future Work

Richer Data Integration:

- Incorporate additional unstructured sources (such as social media, policy texts, migration records) to improve identification of emerging migration signals.

Global Scalability

- Expand the framework to additional countries using globally available demographic and geopolitical datasets.

Near Real-Time / Live Data Integration:

- Integrate near real-time geopolitical and news data to enable dynamic updates of scenario drivers.

References

Kolambe, M., & Arora, S. (2024). Forecasting the Future: A Comprehensive Review of Time Series Prediction Techniques. *Journal of Electrical Systems*. https://doi.org/10.52783/jes.1478

Obradović, S. L., & Matović, S. (n.d.). *Demographic future of municipalities in Serbia through the lens of space-time pattern mining*. https://doi.org/10.59954/ppycdsp2024.33

Newsham, N., Kumar, P., Cabrera-Arnau, C., & Rowe, F. (2024). *A Long Short-Term Memory Forecast of sub-National Population Change Across Europe*. https://doi.org/10.31235/osf.io/9azqy

Pulami, M. J., & Nepal, S. (2024). *The Potential Impact of Demographic Changes in Geopolitical Landscape*. https://doi.org/10.3126/ajps.v3i1.61830

Ketaren, A., Ilham, I., Rangkuty, R. P., Husen, M., & Maliati, N. (2023). Harmonizing Diversity: Reviewing Multicultural Awareness and Political Policy. *Malikussaleh Social and Political Reviews*. https://doi.org/10.29103/mspr.v4i2.10847

Barna, I. (2024). Socio-economic factors of ethnic polymorphism in society. *Наукові Записки Тернопільського Національного Педагогічного Університету Імені Володимира Гнатюка*, *57*(2), 22–29. https://doi.org/10.25128/2519-4577.24.2.3

Drakaki, M., & Tzionas, P. (2023). An Investigation of Time Series Models for Forecasting Mixed Migration Flows: Focusing in Germany. Operations Research Forum, 4(2), 1–11. https://doi.org/10.1007/s43069-023-00212-9

Muttarak, R. (2021). Applying Concepts and Tools in Demography for Estimating, Analyzing, and Forecasting Forced Migration. 9(3), 182–196. https://doi.org/10.1177/23315024211042850

Iliopoulos, I., Kopalidis, N., Stavropoulos, G., & Tzovaras, D. (2022). National press monitoring using Natural Language Processing as an early warning signal for prediction of asylum applications flows in Europe. 4802–4806. https://doi.org/10.1109/BigData55660.2022.10020278

Immigration by Country 2025. (2025-07-01). World Population Review. https://worldpopulationreview.com/country-rankings/immigration-by-country

Thank You